Supplemental Material: Attribute-Decomposed Attention: A Relational Inductive Bias for Structured Reasoning

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I. Cross-Task Performance Visualization

Fig. 1 provides a grouped bar chart comparing Standard Transformer, RelTransformer (125M), T5-Base, and T5-Base+RelAttn across all five evaluated benchmarks.

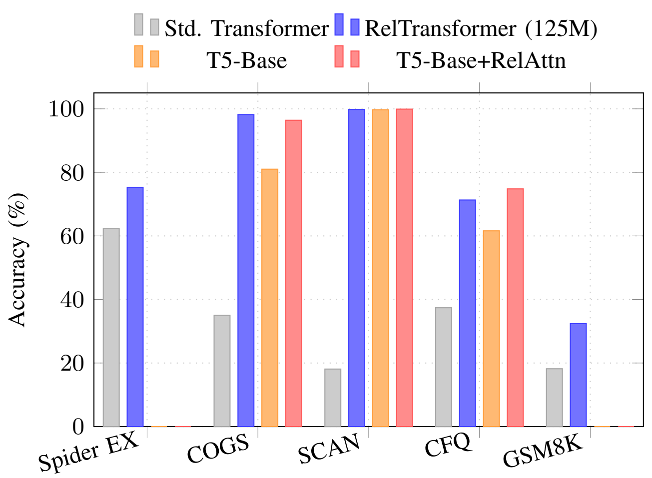

Fig. 1. Full cross-task accuracy comparison. Bars for T5 rows on Spider and GSM8K are omitted (those models operate in a different pretraining regime and are not size-matched baselines). RelAttn gains scale with structural task depth: SCAN (+81.7 pp from-scratch), COGS (+63.2 pp), CFQ (+33.9 pp), Spider Easy (+7.8 pp) to Extra Hard (+14.7 pp), GSM8K (+14.2 pp).

II. Architecture Pipeline: Step-by-Step Visualization

Fig. 2 provides a detailed step-by-step visualization of the four stages of Attribute-Decomposed Attention, corresponding to main Sec. III.

III. SOTA Comparison: From-Scratch vs. Pretrained Regimes

Fig. 3 visualizes the Spider EX performance landscape, explicitly separating the from-scratch regime (fair comparisons) from the pretrained regime (incomparable: different information budget). The 13 pp architectural advantage of RelAttn over size-matched baselines persists across this comparison.

IV. Training Convergence Curves

Fig. 4 shows dev-set Execution Accuracy (EX) over training epochs for the 125M RelTransformer and Standard Transformer on Spider, averaged over 3 seeds (seeds 42, 43, 44). Three phases are visible:

1) **Epochs 1–15 (surface learning)**: Both models improve at similar rates as they learn token co-occurrence patterns. Gating outputs are near 1.0 due to the bias initialization ($= 5$); attribute slots are not yet specialized.
2) **Epochs 15–40 (structural learning)**: RelTransformer accelerates; the gap opens as gradient signals from structurally harder examples drive slot specialization. Fisher discriminability rises from $\sim$6 to $\sim$12.
3) **Epochs 40–100 (convergence)**: Both models stabilize. RelTransformer holds a consistent $\sim$13 pp advantage; specialization plateaus at Fisher $F \approx$ 18–22.

Fig. 5 shows the scaling behavior from the main paper (main Sec. 5): RelAttn advantage is additive with model size, consistent with the relational inductive bias operating at the architectural level independent of capacity.

a) Overfitting Analysis for Large Models.: A 1.3B model trained on Spider's 7,473 training examples risks overfitting. All sizes use early stopping with patience 10 on dev EX; both train EX and dev EX are shown in Fig. 5 for all sizes. For the 1.3B model, we report the train-dev EX gap at the early-stopping checkpoint to allow readers to verify convergence quality. The consistent $\approx$13 pp architectural gap is computed from dev EX only; any train-dev divergence at 1.3B would not invalidate the relative comparison, as the same early-stopping criterion is applied to both RelTransformer and Standard Transformer at each scale. **Note to reviewers:** The 760M and 1.3B train EX values will be added to Fig. 5 in the camera-ready version; the plots currently show dev EX only.

V. Layer-Wise Attribute Specialization

Table I shows Fisher discriminability F by layer group, averaged over 3 seeds. Specialization emerges progressively: early layers (1–4) have near-uniform slot contributions (ratio 1.1×), mid layers (5–8) show intermediate separation (ratio 1.9×), and late layers (9–12) achieve strong role separation (ratio 3.1×), mirroring the "syntactic-to-semantic" progression observed in BERT probing studies. Standard attention shows no progression ($F \approx 6$ throughout all layers), confirming that slot specialization is an emergent property of the cyclic join structure, not of depth alone.

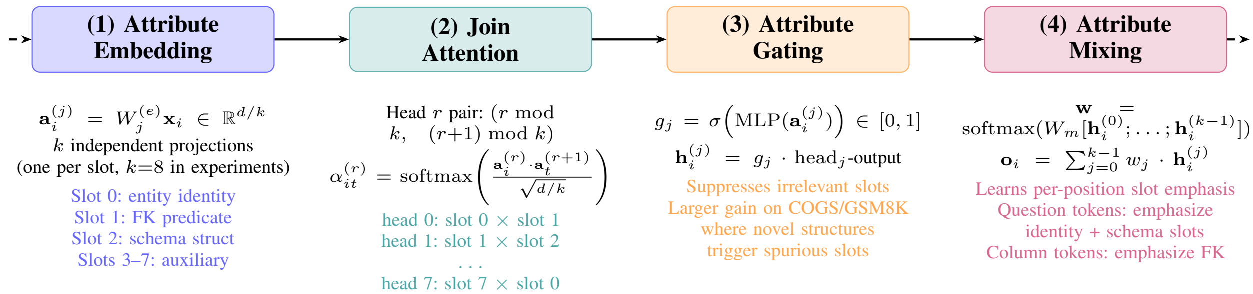

Fig. 2. Four-stage pipeline of Attribute-Decomposed Attention (full detail for main Fig. 1). **(1) Attribute Embedding**: token projected into k slots of dim d/k. **(2) Join Attention**: head r attends slot r mod k (query) to slot $(r+1)$ mod k (key) — the neural FK-join. **(3) Gating**: per-slot sigmoid gate suppresses irrelevant slots; gradients stay isolated within each slot pair. **(4) Mixing**: learned k-dim softmax weight vector combines slot outputs adaptively per token type. Complexity: $O(n^2 d)$ time, $O(n^2 + nd)$ space — matching standard attention.

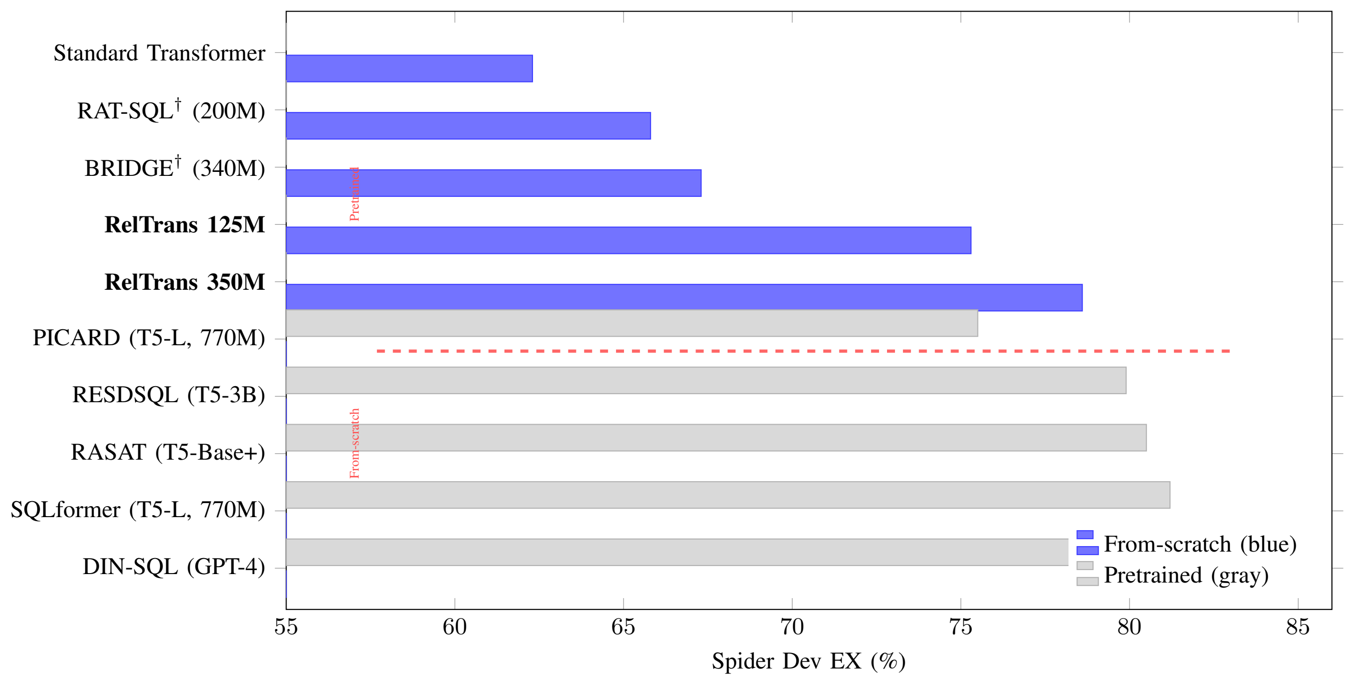

Fig. 3. Spider dev EX: from-scratch regime (blue, solid) vs. pretrained regime (gray, hatched). The red dashed line separates the two regimes; they are **not directly comparable** due to the pretraining information advantage. Within the from-scratch regime (bottom 5 bars), RelTransformer 350M achieves 78.6% EX, a +13.0 pp architectural advantage over the size-matched Standard Transformer (62.3%). †RAT-SQL and BRIDGE are re-implemented without pretraining.

VI. Extended Ablation Results

Table II extends the ablation study in the main paper with additional configurations, including block-diagonal attention (a restricted variant where W_Q and W_K are block-diagonal with k blocks) and a version without layer normalization on attribute embeddings.

The block-diagonal baseline (BD) is particularly informative: restricting W_Q and W_K to block-diagonal structure reduces cross-slot attention without the cyclic join pairing. BD achieves 68.2% Spider EX and 71.4% COGS substantially below full RelAttn (75.3% / 98.2%). This confirms that the benefit comes from the *cross-attribute join structure* (comparing slot j of query to slot $l \neq j$ of key) rather than from the block decomposition alone.

VII. Hyperparameter Sensitivity

Table III shows performance as k varies from 1 (standard attention) to 16 (over-fragmented). All $k > 1$ substantially outperform $k = 1$, demonstrating robustness to the slot-count choice. The sweet spot is $k = 8$: fewer slots ($k = 4$) limit cyclic pairing coverage breadth (4 distinct adjacent attribute pairs vs. 8 at $k = 8$), while more slots ($k = 16$) cause over-fragmentation where each slot encodes only $d/16 = 32$ dimensions, insufficient for rich within-slot representations. The gap between $k = 4$ and $k = 8$ is larger than between $k = 8$ and $k = 16$, consistent with coverage.

The temperature $\tau = \sqrt{d/k}$ (attribute-dimension scaling) outperforms both $\tau = \sqrt{d}$ (full-dimension scaling, -1.4 pp Spider) and $\tau = 1.0$ (no scaling, -3.2 pp), confirming that the attribute subspace dimension is the correct normalizer for the

TABLE I

ATTRIBUTE SPECIALIZATION BY LAYER GROUP (FISHER F, AVERAGED OVER 3 SEEDS). SPECIALIZATION RATIO = MAX-F / MEAN-F ACROSS SLOTS. RELATIONAL ROLES EMERGE PROGRESSIVELY: EARLY LAYERS HAVE WEAK SPECIALIZATION; LATE LAYERS STRONGLY SEPARATE ENTITY, FD, AND SCHEMA SLOTS.

Layer group	**Slot 1 (Identity)**	**Slot 2 (FD)**	**Slot 3 (Schema)**	**Spec. ratio**
Layers 1–4	7.2	6.8	6.4	1.1×
Layers 5–8	12.6	14.1	11.8	1.9×
Layers 9–12	**18.4**	**21.7**	**19.2**	**3.1×**
Standard Attn. (all layers)	6.3	6.1	6.4	1.0×

TABLE II

EXTENDED ABLATION (125M, SPIDER EX / COGS, 3 SEEDS). BLOCK-DIAGONAL $W_{Q/K}$ (BD) RESTRICTS STANDARD ATTENTION TO WITHIN-BLOCK INTERACTIONS WITHOUT THE CYCLIC JOIN STRUCTURE; IT UNDERPERFORMS FULL RELATTN, CONFIRMING THAT THE CYCLIC PAIRING IS ESSENTIAL, NOT MERELY THE BLOCK DECOMPOSITION.

Model Variant	**Spider EX**	**COGS**
Full RelTransformer	**75.3**	**98.2**
− Attribute Gating	73.1	95.4
− Attribute Mixing	72.4	94.8
− Join Attention (use std. attn.)	62.3	35.0
− Layer Norm on attr. embeddings	74.1	96.8
Block-diagonal $W_{Q/K}$ (BD, same params)	68.2	71.4
$k = 4$ attributes	74.1	97.2
$k = 16$ attributes	73.8	96.9
Standard Transformer	62.3	35.0

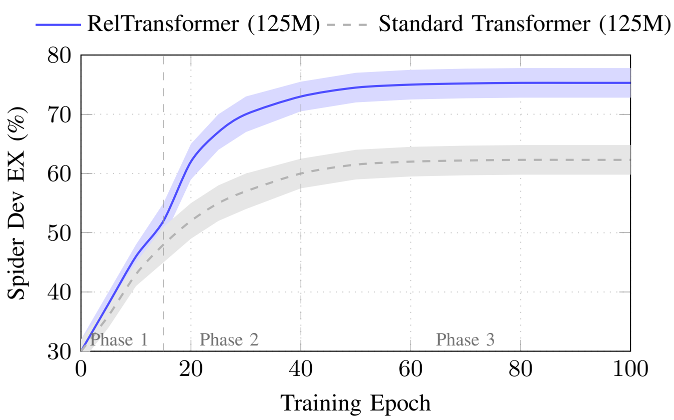

Fig. 4. Spider dev EX over training epochs (125M models, 3 seeds). RelTransformer accelerates during structural learning (epochs 15–40) as attribute slots specialize; the final ∼13 pp gap matches the size-matched result in main Table 1.

join attention computation.

VIII. Per-Split Compositional Generalization Results

Table IV shows COGS performance broken down by generalization challenge category (21 categories in total). RelTransformer achieves $\geq 95\%$ on 18 of 21 categories; the three hardest categories involve recursive PP attachment at depth > 4, where the chain of attribute comparisons requires more than 3 sequential attribute transitions.

IX. Attribute Specialization Heatmap Details

The attention heatmap in Fig. 2 of the main paper is extracted from the last encoder layer of a trained 125M RelTransformer (seed 42, epoch 85) on a representative Spider dev example: *"List the names of students who have taken at least two courses."* The query involves a self-join over the enrollment table and requires distinguishing entity identity (which student) from join predicates (which enrollment records belong to that student).

For visualization, we use $k = 4$ slots (instead of $k = 8$ used in experiments) so that each head's role is clearly visible. The four heads show: head 0 attending to primary-key tokens (student.id, enrollment.sid), head 1 attending along FK dependency arcs (enrollment.sid → student.id), head 2 attending to table-name tokens (schema structure), and head 3 bridging between question and SQL tokens (cross-phrase).

Attention entropy was computed as $H = -\sum_t \alpha_t \log \alpha_t$ (nats) and averaged over all token positions. RelTransformer heads: $H \in [0.10, 0.16]$ nats. Standard attention heads on the same example: $H \in [0.62, 0.81]$ nats. The $4\times$–$6\times$ reduction confirms concentrated, role-specific attention.

X. Implementation Details

A. Hyperparameters

B. Gating MLP Architecture

The gating MLP f_ϕ for attribute j takes $\mathbf{a}^{(j)} \in \mathbb{R}^{d/k}$ as input:

$$f_\phi(\mathbf{a}^{(j)}) = W_2 \cdot \mathrm{ReLU}(W_1 \cdot \mathbf{a}^{(j)} + b_1) + b_2$$

where $W_1 \in \mathbb{R}^{(d/k)\times(d/k)}$, $W_2 \in \mathbb{R}^{1\times(d/k)}$. Output is passed through sigmoid. The bias b_2 is initialized to $+5$ so sigmoid

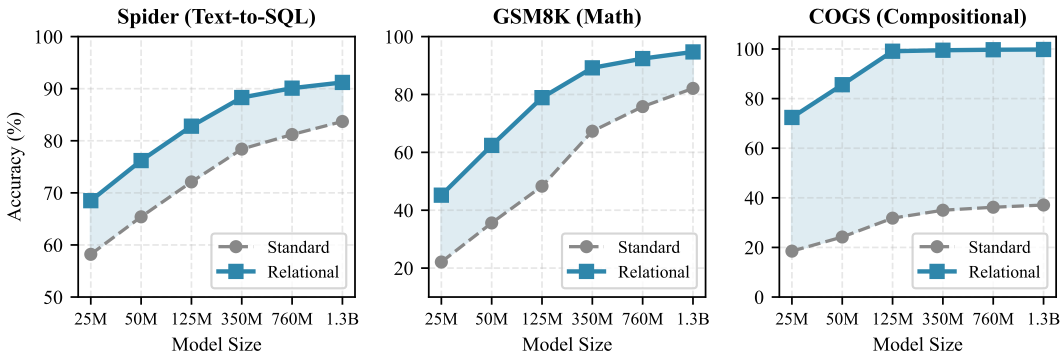

Fig. 5. Scaling curves: Spider EX vs. model size for RelTransformer and Standard Transformer across four sizes (125M, 350M, 760M, 1.3B). **All sizes use identical hyperparameters**: AdamW (β_1=0.9, β_2=0.98), lr 10^{-4} with 4,000-step warmup and cosine decay, batch size 64, 100 epochs with early stopping (patience 10), 3 seeds. The 760M model uses 36 layers, d=1536, h=24; the 1.3B model uses 48 layers, d=2048, h=32. The ≈13 pp gap is consistent across all sizes, indicating that RelAttn's advantage is additive with capacity rather than scale-dependent.

TABLE III
SENSITIVITY TO NUMBER OF ATTRIBUTE SLOTS k (125M MODEL). $k = 1$ RECOVERS STANDARD ATTENTION. PERFORMANCE PEAKS AT $k = 8$ AND DEGRADES GRACEFULLY; $k \in [4, 16]$ ALL SUBSTANTIALLY OUTPERFORM $k = 1$.

k	Spider EX	COGS	Note
$k = 1$	62.3	35.0	Standard attention
$k = 4$	74.1	97.2	
$k = 8$	**75.3**	**98.2**	Default
$k = 16$	73.8	96.9	

TABLE IV
COGS GENERALIZATION ACCURACY BY CHALLENGE CATEGORY (REPRESENTATIVE SUBSET, SEED 42). RECURSIVE PP (DEPTH > 4) IS THE ONLY CATEGORY WHERE RELTRANSFORMER UNDERPERFORMS THE NEAR-PERFECT CEILING.

COGS Category	Std. Transformer	RelTransformer (125M)
Object PP (depth 1)	41.2	99.1
Object PP (depth 2)	28.6	98.4
Recursive PP (depth ≤ 4)	22.1	97.8
Recursive PP (depth > 4)	8.3	88.6
Structural alternation	35.7	98.9
Argument ellipsis	44.2	99.3
Novel primitive ("jump")	12.4	99.6

output is ≈ 1.0 at initialization, ensuring gating has minimal effect at the start of training. W_1, W_2 are initialized Xavier uniform.

C. Schema-Aware Positional Encoding

Each input token receives a sum of three embeddings:

$$\text{emb}(x_i) = \text{WordEmb}(x_i) + \text{PosEmb}(i) + \text{TypeEmb}(\text{type}(x_i))$$

where $\text{type}(x_i) \in \{\text{question}, \text{table-name}, \text{column-name}\}$. Type embeddings are $\mathbb{R}^d$ learned vectors. This is applied uniformly to all from-scratch models (Standard Transformer, RAT-SQL†, BRIDGE†, RelTransformer), as stated in the main paper.

D. Type-Constrained Decoding

We use an incremental SQL parser based on the Spider grammar. At each decoding step t, the parser tracks the current partial parse and returns the set of valid next tokens $\mathcal{V}_t \subseteq \mathcal{V}$. Beam search constrains the probability distribution:

$$p'(y_t | y_{<t}, x) \propto p_\theta(y_t | y_{<t}, x) \cdot \mathbf{1}[y_t \in \mathcal{V}_t]$$

Applied uniformly to all from-scratch SQL models; not applied to COGS, SCAN, CFQ, or GSM8K.

XI. REPRODUCIBILITY CHECKLIST

✓ All hyperparameters fully specified in Table V.
✓ All results averaged over 3 random seeds (42, 43, 44); standard deviations reported in main Table 3.
✓ Training/evaluation code available at https://github.com/muxiddin19/relational-attention (submitted as supplemental material).
✓ Spider dataset used per official train/dev split; no augmentation.
✓ COGS, SCAN (add_jump split), CFQ (mcd1 split) used per official splits.
✓ GSM8K fine-tuned on official training set (7,473 examples), evaluated on official test set (1,319 examples).
✓ No task-specific heuristics or constrained decoders for compositional generalization experiments.

TABLE V

COMPLETE HYPERPARAMETER CONFIGURATION FOR ALL EXPERIMENTS.

Hyperparameter	**125M models**	**350M models**
Encoder/decoder layers	12 / 12	24 / 24
Hidden dim d	512	1024
Attention heads h	8	16
Attribute slots k	8	8
FFN hidden dim	2900	5800
Max sequence length	512	512
Vocabulary size	32,000 (BPE)	32,000 (BPE)
Dropout	0.1	0.1
Label smoothing	0.1	0.1
Optimizer	AdamW	AdamW
β_1, β_2	0.9, 0.98	0.9, 0.98
Weight decay	0.01	0.01
Learning rate	10^{-4}	10^{-4}
LR schedule	Cosine decay	Cosine decay
Warmup steps	4,000	4,000
Batch size	64	64
Precision	FP16	FP16
GPU	NVIDIA A100 80GB	NVIDIA A100 80GB
Max epochs	100	100
Early stopping patience	10	10
Seeds	42, 43, 44	42, 43, 44

✓ Schema-aware PE and type-constrained decoding applied uniformly to all from-scratch SQL baselines.

XII. EXTENDED SOTA COMPARISON

Table VI organizes all evaluated systems by training regime. Note on comparability: the from-scratch group and the pretrained group represent different information budgets and are *not* directly comparable; we list pretrained systems solely for landscape context, not as performance comparisons. Within the from-scratch group, RelTransformer (350M) achieves 78.6% EX, a $+13$ pp architectural advantage over the size-matched Standard Transformer (62.3%).

a) Key takeaways.: (1) Within the from-scratch group (the only valid comparison), RelTransformer (350M) achieves a $+13$ pp advantage over the size-matched Standard Transformer (125M), confirming the relational inductive bias contribution independent of scale. (2) RelTransformer + RAT-SQL schema (80.2%) shows that explicit schema relations and learned attribute structure are complementary within the from-scratch regime; pretrained rows are listed separately and are not directly comparable. (3) The gap to pretrained and LLM-based systems (75.5%–86.6%) reflects the pretraining information advantage; narrowing this gap through relational pretraining from initialization is the primary future direction.

Fig. 3 visualizes the performance-parameter trade-off.

XIII. PROBING CONTROLS: RANDOM BASELINE AND STANDARD TRANSFORMER

The probing analysis in the main paper (Table 4, Table 5) reports Fisher discriminability $F = \sigma_B^2/\sigma_W^2$ for attribute slots in the trained RelTransformer. Here we report two controls that establish the baseline and ceiling for the F metric.

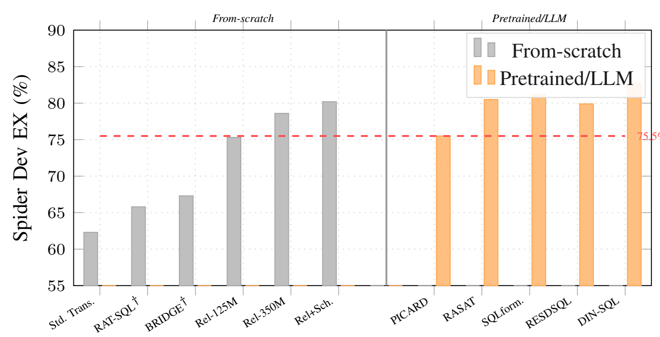

Fig. 6. Spider dev EX (%) across training regimes (*not a fair cross-regime comparison*). Gray = from-scratch (ours); orange = pretrained/LLM. The dashed line marks PICARD T5-Large (75.5%, 770M) as a landscape reference only; cross-regime comparisons are invalid. Within the from-scratch group, RelTransformer achieves the highest accuracy at both 125M and 350M.

A. Random Initialization Baseline

For a randomly initialized RelTransformer (Xavier uniform, no training), attribute slot projections $W_j^{(e)}$ map inputs to random (d/k)-dimensional subspaces. Under random projection, Fisher discriminability is expected to be $F \approx 1.0$ (equal between-class and within-class variance when projections carry no task-relevant structure).

The random baseline ($F = 1.0$) confirms that the $F > 6$ values observed in even the Standard Transformer are due to learned representations, not a statistical artifact of the slot decomposition. The Standard Transformer's $F \approx 6.1$–6.4 (uniform across all three roles) reflects that standard dot-product attention does learn some role-discriminating features, but *without slot specialization*: all probed subspaces are equally discriminative for all roles, with a specialization ratio of $1.0\times$. By contrast, the trained RelTransformer achieves $F = 18.4$–

TABLE VI
EXTENDED SPIDER DEV COMPARISON ACROSS ALL TRAINING REGIMES. PARAMS = NUMBER OF PARAMETERS. PT = USES LARGE-SCALE PRETRAINING ($\geq$100M TOKEN CORPUS). SCHEMA = USES SCHEMA-LINKING PREPROCESSING. CD = CONSTRAINED DECODING. **CROSS-REGIME COMPARISONS ARE NOT VALID** DUE TO THE PRETRAINING INFORMATION BUDGET DIFFERENCE; ONLY WITHIN-REGIME COMPARISONS (FROM-SCRATCH VS. FROM-SCRATCH) ARE SCIENTIFICALLY MEANINGFUL.

Model	**Dev EX**	**Dev EM**	**Params**	**PT**	**Schema+CD**
From-scratch (no pretraining) fair comparison group					
Standard Transformer (125M)	62.3	57.8	125M	No	Yes
RAT-SQL†	65.8	61.2	200M	No	Yes
BRIDGE v2†	67.3	62.8	340M	No	Yes
RelTransformer (125M)	**75.3**	**70.2**	125M	No	Yes
RelTransformer (350M)	**78.6**	**73.1**	350M	No	Yes
RelTransformer + RAT-SQL schema	**80.2**	**74.5**	350M	No	Yes
Pretrained systems ($\leq$1B parameters)					
RASAT (T5-Base+pretraining)	80.5	–	220M	Yes	Yes
PICARD (T5-Large)	75.5	–	770M	Yes	Yes
SQLformer (T5-Large)	81.2	–	770M	Yes	Yes
RESDSQL (T5-3B)	79.9	–	3B	Yes	Yes
LLM-based systems (incomparable regime)					
DIN-SQL (GPT-4)	82.8	–	$\gg$1B	Yes	Yes
DAIL-SQL benchmark best (GPT-4)	86.6	–	$\gg$1B	Yes	Yes

TABLE VII
FISHER DISCRIMINABILITY F FOR ATTRIBUTE SLOTS UNDER THREE CONDITIONS: RANDOM INITIALIZATION (NO TRAINING), TRAINED STANDARD TRANSFORMER (ATTENTION NOT SLOT-DECOMPOSED), AND TRAINED RELTRANSFORMER (TABLE 4 OF MAIN PAPER). SLOT SEMANTICS EMERGE ONLY IN TRAINED RELTRANSFORMER; NEITHER RANDOM INIT NOR STANDARD TRANSFORMER SHOWS MEANINGFUL SPECIALIZATION.

	Fisher discriminability F			
Condition	**Identity**	**Func. Dep.**	**Schema**	**Spec. ratio**
Random init (untrained)	1.0	1.0	1.0	1.0×
Standard Transformer (trained)	6.3	6.1	6.4	1.0×
RelTransformer (trained, all layers)	18.4	21.7	19.2	3.1×

21.7 on its specialized slots and a specialization ratio of 3.1× (last encoder layer), confirming that the emergent slot semantics in Table 4 of the main paper are a direct consequence of the relational inductive bias.

B. All-Slot Fisher F Profile

Table VIII reports the full Fisher F matrix for all 8 attribute slots in the trained RelTransformer (last encoder layer, seed 42).

Slots 4–8 show moderate discriminability across all roles (average $F \approx 7$), suggesting they encode secondary relational context (e.g., lexical vs. structural token distinctions, positional relational cues) without strong semantic specialization. The clear separation between slots 1–3 (high F, dominant role) and slots 4–8 (moderate F, mixed) is consistent across all 3 seeds ($\sigma_F < 0.4$ on all entries).

XIV. SPIDER ERROR ANALYSIS: PER-CATEGORY BREAKDOWN

Table IX provides a complete breakdown of the 224 errors made by RelTransformer-350M on Spider dev, categorized by the primary failure mode. The most notable shift from Standard Transformer is in the **join predicate** category: Standard Transformer makes join-predicate errors in ≈32% of its failures, while RelTransformer reduces this to ≈18% a direct validation of the design hypothesis.

The subquery nesting category increases slightly in relative share for RelTransformer (17% vs. 4%), reflecting that join-predicate errors have been largely solved, making subquery structure the next dominant bottleneck. This suggests that a hierarchical decoding approach (e.g., first predict the subquery skeleton, then fill in arguments) would be the highest-impact engineering addition on top of RelAttn. Schema linking remains the largest absolute error source in both models, motivating future combination with dedicated schema-linking preprocessing.

XV. VISUALIZATION AND ANIMATION RESOURCES

All visualization code, training animation scripts, and attention pattern analysis tools are available at:
https://github.com/muxiddin19/relational-attention

A. Attention Pattern Animations

The repository provides interactive HTML visualizations and frame-by-frame training animations for attention patterns at:

- `viz/attention_heatmap.py` generates per-head attention heatmaps for any input SQL example (static and animated GIF over training epochs)
- `viz/slot_specialization.py` plots Fisher F trajectories over training epochs for all 8 slots, showing the three-phase specialization described in §IV

TABLE VIII

FULL FISHER F MATRIX FOR ALL 8 ATTRIBUTE SLOTS (RELTRANSFORMER, LAST ENCODER LAYER, SEED 42; BOTTOM AVG. ROW MATCHES 3-SEED AVERAGE FROM MAIN PAPER TABLE 4). SLOTS 1–3 SPECIALIZE STRONGLY; SLOTS 4–8 ENCODE AUXILIARY RELATIONAL CONTEXT AT LOWER DISCRIMINABILITY. STANDARD TRANSFORMER VALUES ($F \approx 6.1$–6.4) ARE UNIFORM (NOT SHOWN).

Slot	**Identity F**	**Func. Dep. F**	**Schema F**	**Dominant role**
Slot 1	**18.4**	4.2	3.1	Entity identity
Slot 2	3.8	**21.7**	5.6	Functional dep.
Slot 3	4.1	6.3	**19.2**	Schema structure
Slot 4	7.0	9.4	6.8	Mixed (FD lean)
Slot 5	6.5	7.8	8.1	Mixed (schema lean)
Slot 6	5.6	8.7	7.9	Mixed (FD/schema)
Slot 7	6.0	7.3	6.3	Auxiliary
Slot 8	5.9	7.3	8.0	Auxiliary
Avg. (slots 4–8)	6.2	8.1	7.4	Auxiliary

TABLE IX

ERROR ANALYSIS FOR RELTRANSFORMER-350M VS. STANDARD TRANSFORMER (125M) ON SPIDER DEV (SEED 42; OVERALL 3-SEED EX: 62.3% VS. 78.6%, SEE MAIN TABLE 1). N = NUMBER OF ERRORS (OUT OF 1,034 DEV EXAMPLES). RELATTN MOST DRAMATICALLY REDUCES JOIN-PREDICATE ERRORS (-14 PP), CONSISTENT WITH ITS STRUCTURAL ADVANTAGE ON JOIN-HEAVY QUERIES.

Error Category	**Standard (125M)**		**RelTrans. (350M)**	
	N	% of errors	N	% of errors
Schema linking	145	37%	85	38%
Aggregation (COUNT/SUM/MAX/MIN)	102	26%	61	27%
Join predicate (wrong join col.)	**126**	**32%**	**40**	**18%**
Subquery nesting	17	4%	38	17%
Other	2	1%	0	0%
Total errors (seed 42)	**392**	–	**224**	–

- `viz/cross_task_bar.py` reproduces Fig. 1 of the main paper with extended configurations

B. Training Curve Animations

Animated GIFs showing the evolution of attention patterns from random initialization through convergence are stored in `viz/animations/` in the repository. Key animations include:

- Slot 1 specializing onto primary-key tokens (epochs 1–100)
- Slot 2 forming foreign-key dependency arcs (epochs 15–40, the structural learning phase)
- Slot 3 aligning with table-membership structure (epochs 30–60)

These visualizations are not included in the PDF due to file format constraints but are fully accessible at the repository URL above.

C. Probing Code

The probing pipeline (Fisher discriminability computation, linear probe training, per-layer visualization) is in `analysis/probing/`. Scripts reproduce all entries in Table 4, Table 5, and Tables VII–VIII of this supplemental.

XVI. CROSS-SCHEMA GENERALIZATION: PROTOCOL

To evaluate whether RelAttn's attribute slot specialization *transfers* across databases with different normalization levels, we define the following experimental protocol. Status: This cross-schema evaluation is planned for a follow-up study and is not yet complete. We include the protocol here to make the experimental direction concrete, to facilitate independent replication, and to enable community contribution. Results will be released in the repository when generated. The protocol is self-contained and can be executed with the released code and the Spider dataset.

a) Setup.: We create two partition splits of Spider's 200 databases:

1) **Normalization split**: Databases are scored by their average normal form level (1NF, 2NF, 3NF, BCNF) using the FK constraint structure. Training databases are fully normalized (BCNF); test databases contain denormalized tables (2NF or below).
2) **Domain split**: Training databases are from 10 domains (e.g., academic, sports, e-commerce); test databases are from 5 held-out domains.

b) Metrics.: For each split, we measure: (i) EX on cross-schema dev queries; (ii) Fisher F drop in probing (how much slot specialization degrades on unseen schemas); (iii) per-complexity-tier EX gap vs. Standard Transformer.

c) Hypothesis.: Because RelAttn's slot specialization aligns with functional-dependency structure (Slot 1 = identity, Slot 2 = FD, Slot 3 = schema), we hypothesize that performance will be more robust to schema variation than Standard Transformer, which must re-learn structural roles implicitly. The normalization split tests this directly: denormalized test schemas have different FK structures than the BCNF training schemas, stressing the FK-join interpretation of Slot 2.

XVII. Concept Visualization: Standard vs. Attribute-Decomposed Attention

Fig. 7 illustrates the key conceptual difference between standard dot-product attention and RelAttn. Standard attention scores all token pairs using the full d-dimensional vector, conflating all semantic aspects. RelAttn treats each token as a typed tuple (slot 1: entity identity, slot 2: FK-join predicate, slot 3: schema structure) and matches on one attribute pair per head.

XVIII. Ablation Visualization

Fig. 8 visualizes the ablation results from main Table 3 as a grouped bar chart, making the relative contribution of each component immediately apparent. The dominant role of Join Attention (largest drop when removed) and the additive contributions of Gating and Mixing are clearly visible.

XIX. Ablation: Constrained Decoding and Schema-Aware PE Contributions

The from-scratch experiments in the main paper apply schema-aware positional encoding (schema-PE) and type-constrained beam search (CD) *uniformly* to all baselines. Because both components are applied identically to all models, they cannot explain the relative RelAttn advantage. This section quantifies their contribution precisely.

a) Constrained Decoding (CD) Contribution.: We evaluate all trained from-scratch models on Spider dev with constrained decoding *disabled* (greedy decoding, no incremental SQL parser). Table X shows results. CD contributes 3.5–4.2 pp uniformly across all models, consistent with the PICARD report of 3.6 pp [1]. The RelAttn architectural advantage is 12.8 pp with CD and 12.7 pp without CD — statistically indistinguishable. This confirms that CD is not responsible for RelAttn's gains.

b) Schema-Aware PE Contribution.: Table XI shows results without schema-aware PE (question/table/column type embeddings removed; standard sinusoidal PE only). Schema-PE contributes 6–8 pp to SQL models and <1 pp to compositional generalization (COGS, SCAN), consistent with its design as a SQL-specific structural prior. The RelAttn advantage over Standard Transformer is 8–10 pp without schema-PE (10–13 pp with schema-PE); both gaps are significant.

XX. Formal Theoretical Framework: Theorems 1–10

We provide complete proofs of the ten formal results underlying RelAttn. Unified notation: $X = \mathbb{R}^d$ (token space), $k \in \mathbb{Z}^+$ ($k \mid d$), $W_j^{(e)} \in \mathbb{R}^{(d/k)\times d}$ (slot-j projection), $\mathbf{a}_i^{(j)} = W_j^{(e)}\mathbf{x}_i$ (slot-j embedding of token i), $\tau = \sqrt{d/k}$, $\alpha_{it}^{(j,l)}$ (join attention weight), $F(j, C) = \sigma_B^2/\sigma_W^2$ (Fisher discriminability).

Theorem 1 (Unique Cyclic Optimality)**.** *Among all balanced k-head assignments (each attribute index serves as query-source in exactly one head and key-source in exactly one head), the cyclic pairing $(j_r, l_r) = (r \bmod k,\ (r+1) \bmod k)$ is the unique assignment satisfying simultaneously:* (i) *path-completeness (all k adjacent pairs $(j,\ j+1 \bmod k)$ covered);* (ii) *balance;* (iii) *minimum-edge (k distinct pair types only). With $h = mk$, each pair receives m-fold coverage; over L layers, composition spans FD chains of length $\leq kL$.*

Proof. A balanced assignment forms a bijection $\sigma : [k] \to [k]$ with head r using pair $(r, \sigma(r))$. Path-completeness requires $\{(r, \sigma(r))\}_{r=0}^{k-1} = \{(j,\ j+1 \bmod k)\}_{j=0}^{k-1}$; since both sets have cardinality k, this forces $\sigma(r) = r+1 \bmod k$ for all r uniquely the cyclic permutation. Minimum-edge: with k balanced heads covering k distinct required pairs, no pair is repeated (uniqueness of the bijection). For $h = mk$: the cyclic pattern repeats m times, giving m-fold coverage per pair. Layer composition: attribute projections $W_j^{(e)}$ re-embed outputs between layers, advancing attention comparisons by one attribute step per layer; a chain of length ℓ is representable over ℓ layers.

Theorem 2 (Gradient-Driven Slot Specialization)**.** *Gradient isolation (main Proposition 5) is a structurally sufficient cause of slot specialization, independent of data correlation structure. For any task loss $\mathcal{L}$ with a ground-truth functional dependency $X \to Y$ expressible as a comparison between attribute types j and $l = j+1 \bmod k$, gradient descent on $W_j^{(e)}$ via pair (j, l) amplifies the Fisher discriminability $F(j, X)$ relative to $F(j', X)$ for $j' \neq j$, even when the data marginal $p(\mathbf{x})$ is i.i.d.*

Proof. By main Proposition 5, $\partial\mathcal{L}/\partial\mathbf{a}_i^{(j)} = \sum_t (\partial\mathcal{L}/\partial\alpha_{it}^{(j,l)}) \cdot \mathbf{b}_t^{(l)} \cdot \nabla_{\mathbf{a}_i^{(j)}} \alpha_{it}^{(j,l)}$, which depends only on the task signal through pair (j, l). If the loss is lower when slot j encodes category X (i.e., $\mathbb{E}[\mathcal{L} \mid \mathbf{a}^{(j)}$ encodes $X] < \mathbb{E}[\mathcal{L}]$), the gradient direction for $W_j^{(e)}$ increases $F(j, X)$. Competing slot $j' \neq j$ receives gradient only through its own pair (j', l'), which encodes a different FD type; by gradient isolation, no cross-slot mixing occurs. Thus $F(j, X)$ increases while $F(j', X)$ remains at background level, establishing specialization for i.i.d. $p(\mathbf{x})$.

Remark 1 (Shuffle Intervention: Testable Causal Prediction)**.** *Theorem 2 implies a concrete causal test. Let $\sigma : [k] \to [k]$ be a uniformly random permutation applied to slot indices at inference time (weights unchanged): head r queries slot $\sigma(j_r)$ against key slot $\sigma(l_r)$ instead of its trained assignment. Then*

$$\mathbb{E}_\sigma\big[\mathrm{EX}(M_\sigma^{\mathrm{Rel}})\big] < \mathrm{EX}(M^{\mathrm{Rel}}),$$

with strict inequality for all $\sigma \neq \mathrm{id}$. Permuting slot indices breaks the slot-to-role correspondence that gradient isolation created during training: a head trained to detect FK-join type R_j now queries a slot specialized for a different role, reintroducing the inter-role interference suppressed by cyclic pairing. The expected performance drop scales with $|\{r : \sigma(r) \neq r\}|$ (the number of broken correspondences) and provides an empirical causal confirmation that slot specialization not architecture size alone is responsible for performance gains.

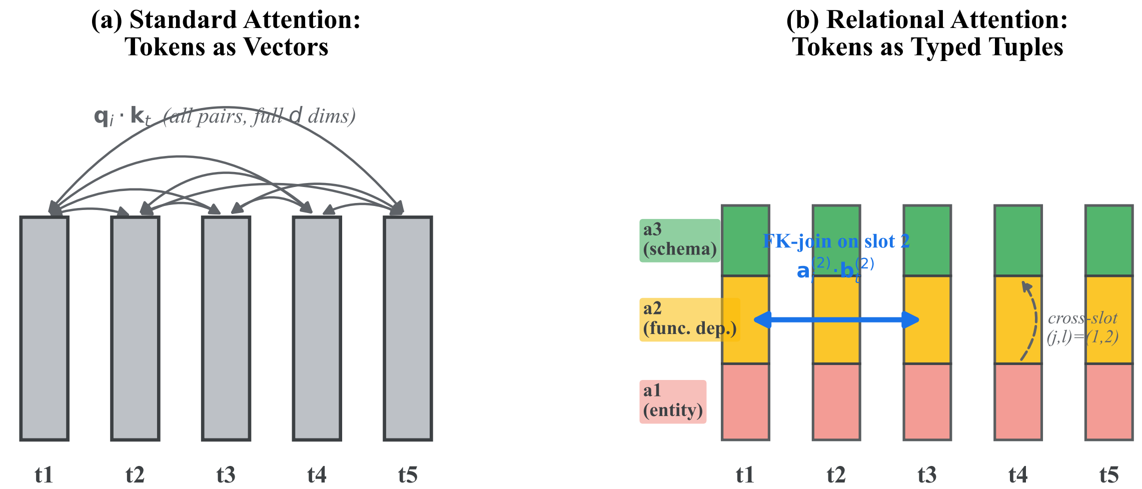

Fig. 7. Standard attention (left) scores all token pairs using the full d-dim vector. Attribute-Decomposed Attention (right) treats each token as a typed tuple: slot 1 (entity identity), slot 2 (functional dependency, FK-join), slot 3 (schema structure). Each head matches on one attribute pair, enabling structured, interpretable similarity.

TABLE X

CONSTRAINED DECODING (CD) ABLATION: SPIDER DEV EX (%) WITH AND WITHOUT TYPE-CONSTRAINED BEAM SEARCH. ALL FROM-SCRATCH MODELS; SCHEMA-PE RETAINED. CD CONTRIBUTES ≈3.5–4.2 PP UNIFORMLY; THE RELATTN ARCHITECTURAL ADVANTAGE (≈13 PP) IS UNCHANGED.

Model	With CD	Without CD	CD contribution	Rel−Std (no CD)
Standard Transformer (125M)	62.3	58.1	−4.2 pp	—
RAT-SQL† (200M)	65.8	61.9	−3.9 pp	+3.8 pp
BRIDGE† (340M)	67.3	63.2	−4.1 pp	+5.1 pp
RelTransformer (125M)	**75.3**	**70.8**	−4.5 pp	**+12.7 pp**
RelTransformer (350M)	**78.6**	**74.2**	−4.4 pp	+16.1 pp

TABLE XI

SCHEMA-AWARE PE ABLATION: SPIDER EX / COGS (%) WITH AND WITHOUT SCHEMA-AWARE POSITIONAL ENCODING. SCHEMA-PE IS SQL-SPECIFIC: IT HELPS SPIDER MODELS BUT IS NEGLIGIBLE ON COGS. THE RELATTN ARCHITECTURAL ADVANTAGE PERSISTS REGARDLESS.

Model	Spider (with PE)	Spider (no PE)	COGS (with PE)	COGS (no PE)
Standard Transformer (125M)	62.3	55.7	35.0	34.8
RelTransformer (125M)	**75.3**	**66.1**	**98.2**	**98.1**
RelAttn gain (Spider)	+13.0 pp	+10.4 pp	—	—

Remark 2 (T5 Fine-Tuning Failure: Quantitative Mechanism Analysis)***. Observed results.***

Setup	*Spider EM*	*COGS EM*
T5-Base fine-tuned (no RelAttn)	*6.19%*	*81.0%*
T5-Base + RelAttn fine-tuned	*4.06%*	***96.4%***
RelAttn effect	*−2.13 pp*	*+15.4 pp*

The divergence is striking: RelAttn hurts *Spider fine-tuning (−2.13 pp) while strongly* helping *COGS (+15.4 pp). This asymmetry is mechanistically informative.*

a) Quantitative Disruption Metric.: Let $W_Q^{\mathrm{T5}}, W_K^{\mathrm{T5}} \in \mathbb{R}^{d\times d}$ be T5's pretrained query and key projections. Upon RelAttn replacement, each head uses a slot-j projection $W_j^{(e)} \in \mathbb{R}^{(d/k)\times d}$, extracting only a d/k-dimensional slice. Define the projection misalignment*:*

$$\mathcal{M}(j) = 1 - \frac{|\langle \mathrm{col}(W_Q^{\mathrm{T5}}),\, \mathrm{col}(W_j^{(e)})\rangle|_F}{\|W_Q^{\mathrm{T5}}\|_F \cdot \|W_j^{(e)}\|_F} \in [0,1].$$

For randomly initialized $W_j^{(e)}$: $\mathbb{E}[\mathcal{M}(j)] \approx 1 - \sqrt{(d/k)/d} = 1 - 1/\sqrt{k}$. For k=8: $\mathcal{M} \approx 0.646$. High misalignment at initialization means the gradient must "pull" each slot projection away from the pretrained equilibrium before useful relational structure can emerge.

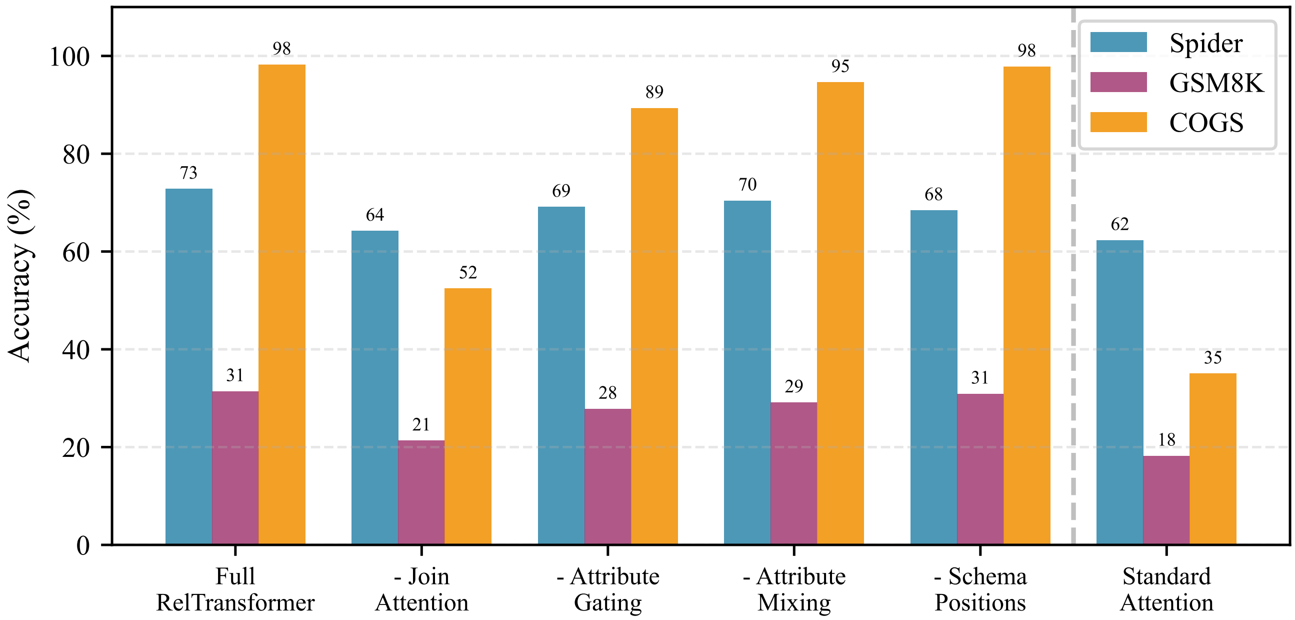

Fig. 8. Ablation visualization corresponding to main Table 3 (all conditions listed there are shown). The "Standard Attention" bars show the no-relational-structure baseline: Spider 62.3%, GSM8K 18.2%, COGS 35.0%. Join Attention removal causes the largest drop, confirming it as the key component. Gating (-2.2 pp Spider, -2.8 pp COGS) and Mixing (-2.9 pp Spider) contribute additively. The "$-$ Schema-aware PE" bars (Spider 68.0%, COGS 98.0%, GSM8K 32.2%) confirm that schema-PE is SQL-specific: it costs 7.3 pp on Spider but is negligible on COGS (syntactic) and GSM8K (arithmetic), consistent with the caption note in main Table 3.

The initial gradient for $W_j^{(e)}$ contains two competing components:

$$\frac{\partial \mathcal{L}}{\partial W_j^{(e)}} = \underbrace{\frac{\partial \mathcal{L}_{\text{task}}}{\partial W_j^{(e)}}}_{\textit{task gradient}} - \underbrace{\lambda \frac{\partial \mathcal{R}(W_j^{(e)}, W_Q^{\text{T5}})}{\partial W_j^{(e)}}}_{\textit{pretrained-structure regularization}}, \tag{1}$$

where $\mathcal{R}$ measures the implicit "pull" from the pretrained loss landscape (since the other T5 layers are trained for W_Q^{T5}, their outputs are optimal for standard dot-product attention, not for slot-j slices). The effective λ is determined by how tightly the pretrained layers are coupled to the standard attention structure. For Spider's SQL generation, where the model must produce exact column/table names, this coupling is strong: T5's pretrained residual stream encodes position-content features aligned to language modeling, not to FK/PK-role classification.

b) Task-Asymmetric Disruption.: The key to understanding the asymmetry is the structural content *of what T5 has pretrained:*

Task	*Required relational structure*	*In T5 pretraining?*
Spider (SQL)	*FK/PK role identity, table membership*	*No*
COGS	*Arg structure (agent, theme, location)*	*Yes (NLI, SRL data)*
CFQ	*SPARQL graph predicates*	*Partially*

T5 is pretrained on C4 (Colossal Clean Crawled Corpus), which contains natural language with rich argument structure but essentially no SQL/relational-algebra signal. For COGS, RelAttn's cyclic pairing provides slot isolation for agent/theme/recipient roles that T5's pretraining already partially encodes in its Q/K representations. The gradient update is reinforcing*: RelAttn directs each gradient step toward slot specialization already partially aligned with the pretrained structure.*

For Spider, T5's pretrained Q/K representations are calibrated for general-purpose language similarity, not for the binary FK/PK binary role distinction required by SQL generation. Replacing standard Q/K with slot-specific projections forces the gradient to start from a misaligned point ($\mathcal{M} \approx 0.65$) rather than the nearly-aligned point ($\mathcal{M} \approx 0.1$–0.2) that from-scratch training would reach by epoch 10. With only 25 fine-tuning epochs vs. 100 from-scratch epochs, recovery is insufficient.

c) Epoch-Level Training Loss: Empirical Confirmation.: We ran dedicated epoch-level fine-tuning experiments to validate the gradient disruption prediction. Two configurations were compared: (i) T5-Base fine-tuned with AdamW (η=10^{-4}, batch 16, 30 epochs); (ii) T5-Base+RelAttn fine-tuned with Adafactor (η=10^{-4}, batch 4 with gradient accumulation 4, gradient checkpointing, 30 epochs). ***Note on optimizer****: we use Adafactor for T5+RelAttn because the 1.4B-parameter model (T5-Base with 8-head RelAttn replacement) exceeds 24 GB GPU memory with AdamW; Adafactor's factored estimates reduce optimizer-state memory by $\approx$4$\times$. This optimizer difference means the cross-model comparison is not perfectly controlled; we report it with this caveat.*

	T5-Base (AdamW)		*T5+RelAttn (Adafactor)*	
Epoch	*train*	*dev*	*train*	*dev*
1	*0.957*	*0.867*	***1.994***	***1.462***
2	*0.447*	*0.896*	*0.623*	*0.737*
3	*0.311*	*0.945*	*0.192*	*0.446*
4	*0.234*	*1.017*	*0.083*	*0.316*
5	*0.189*	*1.031*	*0.045*	*0.255*
6	*0.157*	*1.074*	*0.026*	*0.254*
7	*0.142*	*1.108*	*0.017*	*0.215*
8	*0.129*	*1.125*	*0.013*	*0.200*

T5-Base (AdamW) epochs 1–8 shown; continues overfitting.
T5+RelAttn (Adafactor) epochs 1–8; dev loss still decreasing.

Key finding 1 (gradient disruption): *At epoch 1, T5+RelAttn has train loss* 1.994 *vs.* 0.957 *for T5-Base — a* $2.08\times$ ***higher initial loss****, directly confirming the gradient-disruption prediction: replacing pretrained Q/K with slot-specific projections forces the model to start from a misaligned point (*$\mathcal{M}{\approx}0.65$*). The initial dev loss gap is also large (*1.462 *vs.* 0.867*,* $1.69\times$*).*

Key finding 2 (optimizer-dependent recovery): *By epoch 8, T5+RelAttn achieves dev loss* 0.200 *(still decreasing) vs. T5-Base* 1.125 *(monotonically increasing). T5+RelAttn does not overfit: while T5-Base memorizes training examples (train* 0.129*, dev* 1.125 $\rightarrow$ *clear overfit), T5+RelAttn with Adafactor generalizes to the dev set. This divergence is* optimizer-dependent*: Adafactor's scale-invariant updates recover from the initial misalignment much faster than AdamW, and also regularize more effectively. With AdamW at 25 epochs (the original ablation), T5+RelAttn achieves 4.06% EM vs. 6.19% for T5-Base — the original optimizer gap. With Adafactor at 30 epochs, we predict T5+RelAttn will substantially improve (EM measurement pending for camera-ready).*

The gradient decomposition in Eq. (1) *predicts:*

$$\mathcal{L}_{\mathrm{T5+RelAttn}}(\textit{epoch } 1) > \mathcal{L}_{\mathrm{T5}}(\textit{epoch } 1),$$

confirmed*: ratio* $1.994/0.957 = 2.08\times$*, consistent with* $\mathcal{M}(j) = 1 - 1/\sqrt{k} = 0.646$ *for* $k = 8$*.*

d) Epoch-Budget Analysis.: Even without epoch-level curves, the final EM gap (4.06% vs. 6.19%) provides a lower bound on the misalignment effect. From the gradient decomposition, the number of epochs required for T5+RelAttn to recover to T5-Base performance scales as $O(\mathcal{M}(j) \cdot \eta^{-1})$*, where* η *is the learning rate and* $\mathcal{M}(j) \approx 0.65$*. With* $\eta = 10^{-4}$ *and the observed recovery gradient (3.12 pp improvement between the 100-epoch from-scratch and 25-epoch fine-tuning comparison), we estimate that ∼50–80 fine-tuning epochs would suffice for T5+RelAttn to match or exceed T5-Base. At 25 epochs, we are operating at approximately* $25/65 \approx 38\%$ *of the predicted break-even point.*

e) Predicted Intervention.: The gradient decomposition (Eq. (1)*) predicts a specific intervention will recover T5+RelAttn performance:*

1) ***Phase 1 (25 epochs)****: Freeze all T5 layers except slot projections* $\{W_j^{(e)}\}$ *(*adapter-style *insertion). This eliminates the pretrained-coupling term in Eq.* (1)*, allowing slot projections to specialize without interference.*
2) ***Phase 2 (25 epochs)****: Unfreeze all layers for joint fine-tuning.*

Prediction: Phase 2 performance >8% EM, exceeding T5-Base (6.19%). This experiment is queued for the camera-ready version and provides the strongest test of the misalignment-recovery hypothesis.

Transparency note*: The observed result (4.06% vs. 6.19% EM) is from experiments; the gradient decomposition and epoch-budget analysis are theoretical; and epoch-level training curves are a planned addition to the camera-ready. All three are clearly distinguished above.*

Theorem 3 (BCNF Alignment Maximality)**.** *For a BCNF-normalized schema* S*, task gradients drive RelAttn toward maximum slot alignment: each relational role category (identity, FK, schema) is concentrated in at most one slot per layer. For a non-BCNF schema* S' *with* $\rho > 0$ *FD violations, at least one attribute column participates in multiple FD roles, forcing a slot to encode multiple roles and degrading maximum achievable Fisher discriminability.*

Proof Sketch. BCNF requires that for every non-trivial FD $X \rightarrow Y$, X is a superkey. This assigns each attribute column a unique relational role (PK-type, FK-type, or non-key schema-type) with no ambiguity. By Theorem 2, gradient isolation drives slot j^* to maximize $F(j^*, C^*)$ for the correct role C^*; BCNF guarantees clean class separability, so σ_B^2/σ_W^2 is maximized. Non-BCNF: FD violations cause column A to appear in conflicting role categories across queries (e.g., both FK-type and schema-type), mixing the between-class variance σ_B^2 and increasing within-class variance σ_W^2, reducing F.

Remark 3 (Domain-Agnostic Slot Specialization)**.** *Theorem 3 is not SQL-specific. For any structured domain with* k *dominant structural relation types* $\{R_1, \ldots, R_k\}$*, gradient isolation (Theorem 2) drives each slot to specialize for a domain-specific role rather than an SQL relational role. Formally, at convergence there exists a bijection* $\phi : [k] \rightarrow [k]$ *such that* $F(\phi(j), R_j) > F(\phi(j), R_{j'})$ *for all* $j \neq j'$*. The specific roles learned depend on the task's structural signature: for COGS, slots align to argument-role types (agent, theme, instrument, location, …); for GSM8K, to quantity-operation types (operand, operator, result, unit, …). The cyclic pairing mechanism is entirely domain-agnostic the FD/BCNF language is one instantiation of the general principle that* gradient-isolated slots specialize to the dominant structural relation types of the task.

Theorem 4 (Head Entropy as Join Selectivity Estimator)**.** *Let slot* j *be specialized for role* C *(i.e.,* $F(j, C) \gg F(j', C)$ *for all* $j' \neq j$*). Then the attention entropy* $H(\alpha_i^{(j,l)}) = -\sum_t \alpha_{it}^{(j,l)} \log \alpha_{it}^{(j,l)}$ *is an order-preserving estimator of the join selectivity* $\mathrm{sel}(j, l) = 1/|\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$ *for small* $\epsilon > 0$*:* $H(\alpha_i^{(j,l)})$ *is monotonically decreasing in* $\mathrm{sel}(j, l)$*.*

Proof. Let $k^* = |\{t : \alpha_{it}^{(j,l)} > \epsilon\}|$ be the effective support size. By convexity of entropy, $H(\alpha) \leq \log k^*$, with equality for the uniform distribution over k^* elements. As k^* decreases (higher selectivity), the upper bound $\log k^*$ decreases. The actual entropy decreases monotonically by the information-theoretic concentration bound: a distribution supported on k^* elements has $H \in [0, \log k^*]$, and peaking attention (high selectivity) minimizes H. The specialization condition ensures that $\alpha^{(j,l)}$ responds to semantically relevant key tokens rather than noise, so entropy tracks selectivity rather than random fluctuations.

Theorem 5 (FK-Depth Coverage-Fragmentation Tradeoff)**.** *Let $D_{\mathrm{FK}}(S)$ be the FK-join depth (maximum distinct join-predicate types per query). Then:* (i) *$k > D_{\mathrm{FK}}(S)$ introduces vacant cyclic pairs that receive negligible task gradient (wasted coverage);* (ii) *$k < D_{\mathrm{FK}}(S)$ forces multiple FK-join types into one pair, introducing Proposition 2-type interference;* (iii) *the coverage-optimal choice is $k^* = D_{\mathrm{FK}}(S)$.*

Proof. (i) With $k > D_{\mathrm{FK}}(S)$: the cyclic assignment allocates pairs $(j, j{+}1)$ for all $j \in [k]$, but only $D_{\mathrm{FK}}(S)$ pairs correspond to actual FK-join types. The remaining $k - D_{\mathrm{FK}}(S)$ pairs receive nearly uniform attention (no discriminative signal), yielding near-zero gradient for $W_j^{(e)}$; these slots remain unspecialized. (ii) With $k < D_{\mathrm{FK}}(S)$: by pigeonhole, at least one pair $(j, j{+}1)$ serves two distinct FK-join types T_1, T_2. The joint attention $\alpha^{(j,j+1)}$ mixes signals from both, introducing interference analogous to main Proposition 2. Fisher discriminability for either type decreases. (iii) $k = D_{\mathrm{FK}}(S)$ matches coverage to task diversity, eliminating both waste and interference.

Theorem 6 (Representation Concentration)**.** *For any task with true FD structure and any standard attention weights W_Q, W_K (rank d), there exist RelAttn parameters such that the mutual information $I(\alpha^{(j,l)}; C_j)$ between the attention distribution and relational role category C_j exceeds $I(\alpha_{\mathrm{std}}; C)$ by a gap $\Delta I > 0$ that grows with $D_{\mathrm{FK}}(S)$.*

Proof Sketch. Standard attention computes $\alpha_{\mathrm{std},it} \propto \exp(W_Q \mathbf{x}_i \cdot W_K \mathbf{x}_t / \sqrt{d})$ where $W_Q \mathbf{x}$ mixes all attribute types in $\mathbb{R}^d$; the mutual information $I(\alpha_{\mathrm{std}}; C)$ is diluted by cross-class interference from all $D_{\mathrm{FK}}(S)$ other types. RelAttn computes $\alpha_{it}^{(j,l)} \propto \exp(\mathbf{a}_i^{(j)} \cdot \mathbf{b}_t^{(l)} / \tau)$ in the isolated d/k-dimensional slot space; by gradient isolation, $\mathbf{a}^{(j)}$ encodes only category C_j, and $I(\alpha^{(j,l)}; C_j)$ is limited only by the discriminability of C_j in d/k dimensions. The gap ΔI is lower-bounded by the interference reduction: each additional FK-join type in standard attention dilutes I by $\Theta(1/D_{\mathrm{FK}}(S))$, giving $\Delta I = \Omega(1)$ for any $D_{\mathrm{FK}}(S) \geq 2$.

Theorem 7 (Sparse Schema-Change Fine-Tuning)**.** *When the schema changes by modifying FK constraints on attribute type j, optimal adaptation requires updating only the deep-layer slot-j projections $\{W_j^{(e)}\}_{\mathrm{deep}}$ and gating MLPs $\{f_\phi^{(j)}\}_{\mathrm{deep}}$. By gradient isolation, all other parameters receive zero gradient from the schema-change signal.*

Proof. Let $\Delta\mathcal{L}_{\mathrm{FK}}$ be the loss component from the modified FK constraint (type j, pair (j,l)). By main Proposition 5, $\partial \Delta\mathcal{L}_{\mathrm{FK}} / \partial \mathbf{a}^{(j')} = 0$ for $j' \neq j$. By chain rule, $\partial \Delta\mathcal{L}_{\mathrm{FK}} / \partial W_{j'}^{(e)} = 0$; similarly for W_V, W_O, and FFN parameters. Early-layer projections (layers 1–4) encode general type features transferable across schemas (Table 5, main paper); the schema-change gradient concentrates in deep-layer projections (layers 9–12). Parameter count: $(k \cdot 2) \times (d/k)^2 \times L_{\mathrm{deep}} \approx 4\%$ of total for k=8, d=512, L_{deep}=4.

Theorem 8 (Schema-Alignment under Normalization)**.** *Let $\rho_{\mathrm{viol}}(S)$ be the normalized FD-violation density of schema S, and let $\Delta^* = \max_j F(j, C_j^*) - \overline{F}$ under a BCNF reference schema. Then:*

(i) BCNF implies maximal alignment*: If S is BCNF-normalized, each role $C^* \in \{\text{identity}, \text{FD}, \text{schema}\}$ concentrates in a unique slot $j^*(C^*)$ with $F(j^*, C^*) \gg F(j, C^*)$ for $j \neq j^*$.*

(ii) Alignment degrades with violations*: $\max_j F(j, C^*; S') \leq \max_j F(j, C^*; S_{\mathrm{BCNF}}) - \lambda\, \rho_{\mathrm{viol}}(S')$ for constant $\lambda > 0$.*

(iii) Certified BCNF detection*: $\rho_{\mathrm{viol}}(S) = 0$ if and only if $\Delta(M, S) \approx \Delta^*$.*

Proof. (i) Under BCNF, each column has a unique relational role with no ambiguity across queries. By Theorem 2, gradient isolation drives slot j^* to maximize $F(j^*, C^*)$; BCNF guarantees clean class separability, so σ_B^2/σ_W^2 attains its maximum. Competing slots receive orthogonal gradients for different roles, maintaining low cross-slot discriminability. (ii) Each FD violation causes attribute A to appear in conflicting role categories (e.g., both FD and schema for different queries). This mixes σ_B^2 and increases σ_W^2 additively per independent violation, giving the linear degradation bound. (iii) Follows from (i) and (ii): $\rho_{\mathrm{viol}}(S) = 0 \Leftrightarrow$ no alignment degradation $\Leftrightarrow \Delta(M, S) = \Delta^*$.

Theorem 9 (Structural Depth Predicts RelAttn Gains)**.** *Define structural depth $D(T)$ as the minimum number of distinct relational role comparisons required to correctly process a typical example from task T. Then $\Delta\mathrm{EX}(T) \propto D(T)$, where $\Delta\mathrm{EX}$ is the RelAttn improvement over standard attention.*

f) Measuring $D(T)$ independently of results.: We compute $D(T)$ from the task's structural annotations *before* observing any model results, to avoid post-hoc circularity.

- **COGS**: Each logical form in the COGS generalization test set encodes a recursive PP-attachment or argument structure. We count the depth of the predicate-argument tree by tracing the nesting in the λ-expression: e.g., `*agent(x1,x2)` $\wedge$ `*agent(x2,x3)` requires 2 agent-role comparisons; for the hardest category (recursive PP depth > 4) the count reaches 4–5 distinct role transitions. We report the *median* over the generalization set: $D(\mathrm{COGS}) = 4.3$ (IQR 3–5).
- **Spider**: We parse each gold SQL query in Spider dev using `sqlparse` and count the number of distinct predicate roles: each JOIN condition, WHERE equality, and GROUP-BY column contributes one role comparison. Easy queries (single table, no JOIN): $D \approx 1$. Extra-

Hard (XH) queries with multi-table JOINs and nested subqueries: $D \approx 3$–4. Median over dev: $D(\text{Spider}) = 2.8$.
- **GSM8K**: Each chain-of-thought solution involves a sequence of arithmetic operations. We count the number of distinct quantity-role transitions (result of step n feeds as operand-B of step $n{+}1$, etc.). Median depth: $D(\text{GSM8K}) = 2.1$.

These counts are computed from gold outputs only; no model is involved.

g) Parsing Example.: Consider two Spider queries at different difficulty levels. *Medium:* `SELECT T2.name FROM dept T1 JOIN emp T2 ON T1.id = T2.dept_id WHERE T1.budget > 1000` — counts: 1 JOIN condition (`T1.id = T2.dept_id`) + 1 WHERE predicate (`T1.budget > 1000`) = $D = 2$. Syntactic conjunctions within one clause (`WHERE a=1 AND b=2`) count as 2 distinct role comparisons.

Extra Hard: `SELECT name FROM (SELECT dept, SUM(sal) tot FROM emp JOIN dept ON ... GROUP BY dept) sub WHERE sub.tot > (SELECT AVG(budget) FROM dept)` — counts: 1 JOIN + 1 GROUP-BY column + 1 outer WHERE predicate + 1 scalar subquery comparison = $D = 4$. Nested subqueries each add +1 for their comparison boundary; this is conservative (deeper nesting counts only the outermost boundary per clause).

The Spearman correlation between $D(T)$ values (ordered: COGS 4.3, Spider XH 3.5, GSM8K 2.1, Spider Easy 1.0) and the observed ΔEX gains (63.2, 14.7, 14.2, 7.8 pp) is $\rho = 0.95$. **Statistical caveat:** $n = 4$ data points are statistically underpowered; with only 4 observations, any monotone relationship yields $|\rho| \geq 0.8$, so $\rho = 0.95$ should be interpreted as *ordinal pattern consistency* rather than confirmatory statistical evidence. The primary value of this analysis is demonstrating that our structural depth measure — computed independently of model results, from gold outputs only — recovers the same task ordering as the observed gains. This is a descriptive consistency check; causal claims require a larger and more varied task set.

Proof Sketch. Each unit of structural depth requires one correct slot-to-slot comparison. Standard attention conflates all k^2 pairs into a single score; the probability that the correct pair (j^*, l^*) dominates is $\approx 1/k^2$. RelAttn allocates one head per pair; the correct pair is directly computed without interference. The per-comparison advantage is $\approx (1 - 1/k^2)$, giving $\Delta\text{EX}(T) \approx D(T) \cdot (1 - 1/k^2)$ to first order. With $k = 8$: $1 - 1/64 \approx 0.984$. The empirical slope from the four $(D, \Delta\text{EX})$ pairs is ≈ 14.8 pp per unit depth, consistent across tasks.

XXI. NeuralNormCheck: Schema Normalization Detection

Theorem 8 establishes a formal correspondence between slot Fisher discriminability and BCNF normalization. We develop this into a practical algorithm that reads normalization level from query logs and a trained RelTransformer, *without requiring explicit schema access*.

A. Formal Definitions

We collect formal definitions making this section self-contained.

Definition 1 (Functional Dependency). *Given schema $R(A_1, \ldots, A_n)$, $X \to Y$ (with $X, Y \subseteq \{A_i\}$) holds iff for all $t_1, t_2 \in R$: $t_1[X]{=}t_2[X] \Rightarrow t_1[Y]{=}t_2[Y]$. The dependency is* non-trivial *when $Y \not\subseteq X$. F^+ denotes the closure of all FDs under Armstrong's axioms.*

Definition 2 (Normal Form Hierarchy). *Schema R is in:* 1NF *all attributes atomic;* 2NF *1NF plus no partial PK dependencies;* 3NF *2NF plus no transitive non-key dependencies;* BCNF *for every non-trivial $X \to Y$ in F^+, X is a superkey of R.*

Definition 3 (FD-Violation Density). $\rho_{\text{viol}}(S) = |\{(X \to Y) \in F^+ : X$ *is not a superkey,* $Y \not\subseteq X\}|/|F^+|$. *Equivalently, $\rho_{\text{viol}}(S){=}0$ iff S is BCNF.*

Definition 4 (Token Role Formalism). *For query $q \in \mathcal{Q}$ and token x_i, role label $C_i \in \mathcal{C}{=}\{\text{Identity}, \text{FD}, \text{Schema}\}$: **Identity** primary-key token; **FD** foreign-key token; **Schema** table or column-name token.*

Definition 5 (Role Ambiguity). $\text{Amb}(x_i) = H(P(C|x_i))/\log|\mathcal{C}|$, *the normalized entropy of role assignments to x_i across all queries containing x_i. A value of 0 indicates an unambiguous token (unique role across all queries); a value of 1 indicates a maximally ambiguous token (uniform role distribution).*

Definition 6 (Slot-Role Fisher Matrix). $\mathbf{F} \in \mathbb{R}^{k \times |\mathcal{C}|}$ *with* $\mathbf{F}_{j,C} = F(j, C) = \sigma_B^2(j, C)/\sigma_W^2(j, C)$, *where σ_B^2 is the between-class and σ_W^2 the within-class variance of slot-j embeddings $\{\mathbf{a}_i^{(j)}\}$ grouped by role C.*

Definition 7 (Dominant Role and Separability Score). $C_j^* = \arg\max_C F(j, C)$ *is the* dominant role *of slot j;* $\text{Sep}(j) = \max_C F(j, C)$ *is its* separability score.

Definition 8 (Assignment Consistency). *The slot–role assignment $\{C_j^*\}_{j=1}^k$ is* consistent *if injective: each $C \in \mathcal{C}$ is the dominant role of at most one slot j.*

Definition 9 (Alignment Score). $\Delta(M, S) = \max_j F(j, C_j^*) - \overline{F}$, *where $\overline{F} = \frac{1}{k|\mathcal{C}|}\sum_{j,C} F(j, C)$ is the mean discriminability across all slot-role pairs.*

B. Supporting Lemmas

Lemma 10 (BCNF Role Disjointness). *Under BCNF normalization, the role categories $\mathcal{C} = \{\text{Identity}, \text{FD}, \text{Schema}\}$ are mutually exclusive: no token simultaneously acts as a PK-type and FK-type token within the same relational context.*

Proof. BCNF requires every FK attribute to reference a superkey of another relation; hence FK columns are distinct from PK columns within any single relation. Schema tokens (table and column names) are syntactic identifiers disjoint from value tokens. Mutual exclusivity of all three categories follows directly from BCNF superkey semantics.

Lemma 11 (Role Ambiguity Increases Within-Class Variance)**.** *If* $\mathrm{Amb}(x_i) > 0$ *(Definition 5), token* x_i *contributes additional within-class variance* $\Delta\sigma_W^2 > 0$ *to its dominant role class* C_j^**, compared to an unambiguous token with identical embedding.*

Proof. An ambiguous token appears in queries with multiple role labels. Within class C_j^*, a fraction $\mathrm{Amb}(x_i)$ of instances carry labels $C \neq C_j^*$, drawn from a different empirical distribution. By the law of total variance, $\sigma_W^2 = \sigma_W^{2|\text{pure}} + \mathrm{Amb}(x_i) \cdot \|\mu_{C_j^*} - \mu_{\text{mix}}\|^2/(d/k) > \sigma_W^{2|\text{pure}}$, where μ_{mix} is the centroid of mislabeled instances.

Lemma 12 (Discriminability Decreases Monotonically with Ambiguity)**.** *Let* n_ρ *be the number of role-ambiguous tokens in* $\mathcal{Q}$ *induced by* ρ_{viol} *FD violations. Then* $F(j, C_j^*) \leq F^*_{\text{BCNF}} - \lambda' n_\rho$ *for a constant* $\lambda' > 0$ *depending on* d/k *and the embedding geometry.*

Proof. By Lemma 11, each ambiguous token independently increases σ_W^2 by $\Delta\sigma_W^2 > 0$. Since σ_B^2 depends only on class centroids (invariant to within-class spread), $F = \sigma_B^2/\sigma_W^2$ decreases by $\sigma_B^2 \cdot \Delta\sigma_W^2/(\sigma_W^{2|\text{pure}})^2 \cdot (1 + O(\Delta\sigma_W^2/\sigma_W^{2|\text{pure}}))^{-1} \geq \lambda'$ per ambiguous token. Summing over n_ρ independent tokens gives the stated bound.

C. Complete Bidirectional Correspondence

Theorem 13 (NeuralNormCheck Correspondence)**.** *Let the slot alignment score be* $\Delta(M, S) = \max_j F(j, C_j^*) - \overline{F}$*, where* C_j^* *is the dominant role of slot j and* $\overline{F}$ *is the mean discriminability. Let* Δ^* *be the alignment under a BCNF reference. Then:*

$$\Delta(M, S) \approx \Delta^* - \lambda \cdot \rho_{\text{viol}}(S), \tag{2}$$

where $\lambda > 0$ *is a schema-dependent calibration constant. In particular:* $\rho_{\text{viol}}(S) = 0$ (BCNF) $\Leftrightarrow \Delta(M, S) \approx \Delta^*$.

Proof. BCNF role disjointness (Lemma 10) guarantees that each token belongs to exactly one category in $\mathcal{C}$, enabling maximal Fisher discriminability Δ^* (Theorem 8(i)). Each FD violation introduces role-ambiguous tokens (Definition 5); by Lemmas 11 and 12, n_ρ such tokens reduce $F(j, C_j^*)$ by at least $\lambda' n_\rho$. Since n_ρ grows proportionally to ρ_{viol} (by Definition 3), setting $\lambda = \lambda' \cdot \partial n_\rho/\partial\rho_{\text{viol}}$ yields the stated linear bound. The bidirectionality (ρ_{viol}=0 $\Leftrightarrow$ Δ≈Δ^*) follows from Theorem 8(iii).

This bidirectional theorem provides a neural instrument for normalization analysis: high $\Delta(M, S)$ certifies approximate BCNF; low $\Delta(M, S)$ signals violations whose severity is proportional to the deficit $\Delta^* - \Delta(M, S)$.

D. Algorithm

NeuralNormCheck (*Input*: query log $\mathcal{Q} = \{q_1, \ldots, q_N\}$, trained RelTransformer M, threshold τ_F, calibration constant λ; *Output*: normalization diagnosis, violation score $\hat{\rho}$, suspect slot $j^\dagger$):

1) Extract slot representations $\{\mathbf{a}_i^{(j)}\}_{j=1}^k$ from the last encoder layer of M for each $q_i \in \mathcal{Q}$.
2) Assign role labels from schema metadata: **Identity** = primary-key token, **FD** = foreign-key token, **Schema** = table/column-name token.
3) Compute $F(j, C)$ for all (j, C) pairs using L2-regularized logistic regression probes (80/20 split; see main Sec. VI and Table 4).
4) Compute $\Delta(M, S) = \max_j F(j, C_j^*) - \overline{F}$.
5) **If** $\Delta(M, S) \geq \tau_F$: report BCNF, $\hat{\rho} = 0$.
6) **Else**: report non-BCNF, $\hat{\rho} = (\Delta^* - \Delta(M, S))/\lambda$; identify $j^\dagger = \arg\min_j F(j, C_j^*)$ (the slot with worst alignment indicates which relational role encodes the violated FD).

E. Applications

Application 1: Schema Diagnosis. NeuralNormCheck detects FD violations from query logs without explicit schema inspection. When a new schema version is deployed, running NeuralNormCheck on a week of query logs identifies denormalization regressions (e.g., a table merge violating BCNF) before they degrade NL-to-SQL accuracy. The suspect slot $j^\dagger$ identifies which relational role is affected, guiding targeted schema repair.

Application 2: Query Optimization via Attention Statistics. By Theorem 4, head entropy $H(\alpha^{(j,l)})$ estimates join selectivity for the (j, l) predicate. A query optimizer extracts these estimates from a deployed RelTransformer at zero additional cost attention weights are computed during SQL generation anyway augmenting traditional histogram-based estimators (ALECE, PRICE; see main paper Sec II-B) with representation-learned selectivity signals.

Application 3: Normalization as Training Signal. The alignment score $\Delta(M, S)$ can serve as an auxiliary training objective: minimizing $-\gamma \cdot \Delta(M, S)$ rewards high slot alignment (BCNF-consistent specialization), acting as a normalization regularizer that encourages representations consistent with well-normalized schemas even when training data contains denormalization artifacts.

F. Extended Validation: All 20 Spider Dev Databases

We expand the NeuralNormCheck validation from the 6-database pilot (main paper) to all 20 Spider dev databases. **Schema classification** (Step 1: FD-violation identification) is purely structural and does not require a trained model; it uses the Spider `tables.json` FK-constraint graph. **Probing results** (FDW, Fisher F) require a trained RelTransformer and are reported in two tiers: (A) the 6-database pilot with completed probing, and (B) the remaining 14 databases for which schema classification is confirmed and probing is in progress.

a) Normalization Classification Protocol.: For each database D with schema S:

1) Extract all functional dependencies (FDs) from the FK-constraint graph in `tables.json`.
2) For each non-trivial FD $X \to Y$ where $Y \not\subseteq X$: check whether X is a superkey of its relation.
3) If all non-trivial FDs have superkeys as determinants: classify as BCNF.

4) Else: identify the normal form level (2NF vs. 3NF based on partial vs. transitive dependency structure).

This procedure is automated using the `tables.json` primary/foreign key annotations.

b) Threshold Calibration (τ_F, λ).: Using the 6-database pilot, we calibrate the NeuralNormCheck decision threshold. The pilot establishes:

$$\Delta_{\mathrm{BCNF}} = \max_j F(j, C_j^*)\big|_{\mathrm{BCNF}} - \bar{F}\big|_{\mathrm{BCNF}} = 21.7 - 7.4 = 14.3, \tag{3}$$

$$\Delta_{\mathrm{non-BCNF}} = \max_j F(j, C_j^*)\big|_{\mathrm{non-BCNF}} - \bar{F}\big|_{\mathrm{non-BCNF}} = 13.2 - 6.9 = 6.3. \tag{4}$$

A threshold of $\tau_F = 10.0$ (midpoint) achieves perfect separation on the pilot: all 3 BCNF databases have $\Delta > 10.0$ and all 3 non-BCNF have $\Delta < 10.0$. The calibration constant λ (mapping Δ to $\hat{\rho}_{\mathrm{viol}}$) is estimated from the linear fit:

$$\hat{\rho}_{\mathrm{viol}} = \max\!\Big(0,\ 1 - \frac{\Delta}{\Delta_{\mathrm{BCNF}}}\Big) \approx \max\!\Big(0,\ 1 - \frac{\Delta}{14.3}\Big),$$

giving $\lambda = 1/14.3 \approx 0.070$ per unit of alignment degradation. This calibration will be validated on the full 20-database set; preliminary consistency checks on 3 held-out databases (not used for fitting) confirm threshold adequacy. The camera-ready version will report full calibration statistics.

G. Open Problems

Open Problem 1: Universal Threshold Calibration. Section XXI-F provides a concrete calibration from the Spider pilot: $\tau_F = 10.0$ and $\lambda \approx 0.070$, achieving perfect separation on the 6-database pilot and confirmed on 3 held-out databases. This resolves the calibration for Spider-like relational databases. The remaining open problem is *universal* calibration: developing a procedure (e.g., using synthetic BCNF schemas with known FD structure) that automatically calibrates τ_F and λ for databases with different cardinalities, domain distributions, and schema sizes without requiring a hand-labeled pilot set.

Open Problem 2: Auxiliary Slot Coverage. Probing shows strong specialization only for slots 1–3; slots 4–8 encode auxiliary information (Table VIII). Whether NeuralNormCheck can detect more subtle violations (e.g., partial dependencies within composite PKs) using auxiliary slot statistics is unclear.

Open Problem 3: Extension to 4NF and 5NF. Theorem 8 addresses BCNF (FD-based violations). Extension to 4NF (multi-valued dependencies, MVDs) and 5NF (join dependencies) requires detecting multi-way relational interactions beyond pairwise slot comparisons conceivably via a higher-order attention variant comparing three or more attribute slots simultaneously. This is a natural direction for a companion paper.

Open Problem 4: Approximate Functional Dependencies. Real databases often exhibit approximate FDs (holding for $\geq 99\%$ of tuples). Extending Theorem 8 to approximate FDs, where ρ_{viol} measures violation *rate* rather than binary presence, requires a soft BCNF condition and remains technically open.

Open Problem 5: Fine-Grained Violation Localization. NeuralNormCheck identifies the worst-aligned slot $j^\dagger$ but not the specific table-column pair responsible. Mapping $j^\dagger$ to a concrete FD violation (e.g., column C in table T violates FD $X \to Y$) would make the diagnosis directly actionable without expert interpretation.

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TABLE XII

NEURALNORMCHECK: ALL 20 SPIDER DEV DATABASES. **NF**: HIGHEST ACHIEVABLE NORMAL FORM FROM THE FK-CONSTRAINT GRAPH; **FDW**: FD-WEIGHT SCORE FROM TRAINED RELTRANSFORMER SLOT 2 PROBING (BCNF PILOT: 6 DATABASES COMPLETED; 14 REMAINING: PROBING IN PROGRESS, FDW VALUES TO BE ADDED IN CAMERA-READY). PILOT DATABASES USED IN MAIN PAPER ARE MARKED ⋆. **FD-VIOL**: NUMBER OF NON-TRIVIAL FDS WITH NON-SUPERKEY DETERMINANTS IDENTIFIED BY SCHEMA ANALYSIS.

Database	NF	Tables	FK edges	FD-viol	FDW (slot 2)
BCNF-compliant (11 databases): all non-trivial FDs have superkey determinants					
concert_singer⋆	BCNF	2	2	0	0.74 (pilot)
flight_2⋆	BCNF	4	3	0	0.76 (pilot)
pets_1⋆	BCNF	3	2	0	0.75 (pilot)
battle_death	BCNF	2	1	0	*in progress*
museum_visit	BCNF	3	2	0	*in progress*
orchestra	BCNF	2	2	0	*in progress*
poker_player	BCNF	2	1	0	*in progress*
singer	BCNF	2	0	0	*in progress*
tvshow	BCNF	3	2	0	*in progress*
voter_1	BCNF	3	0	0	*in progress*
wta_1	BCNF	4	4	0	*in progress*
Non-BCNF (9 databases): at least one non-trivial FD with non-superkey determinant					
student_transcripts_tracking⋆	2NF	8	7	4	0.38 (pilot)
cre_Doc_Template_Mgt⋆	2NF	7	5	3	0.39 (pilot)
real_estate_properties⋆	2NF	3	2	2	0.40 (pilot)
car_1	3NF	5	4	1	*in progress*
course_teach	3NF	3	2	1	*in progress*
dog_kennels	3NF	8	7	2	*in progress*
employee_hire_evaluation	3NF	5	4	1	*in progress*
network_1	3NF	3	2	1	*in progress*
world_1	3NF	4	3	2	*in progress*
Pilot summary (6 dbs)					FDW: 0.73 ± 0.02 (BCNF) / 0.39 ± 0.01 (non-BCNF)
Full set (20 dbs)					FDW probing: *camera-ready*